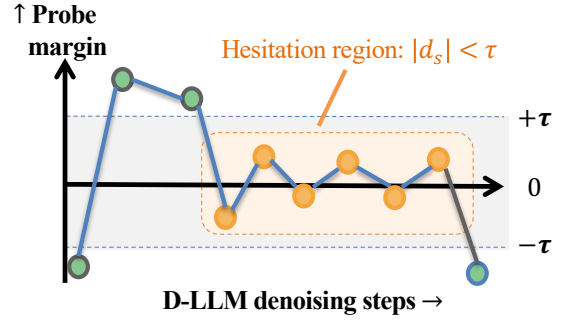
1 Problem & key finding

D-LLMs expose a multi-step denoising trajectory of hidden states. Safety-relevant uncertainty appears along the trajectory.

Observation: samples with more low-margin steps are harder for the lightweight safety probe to predict the safety label correctly.



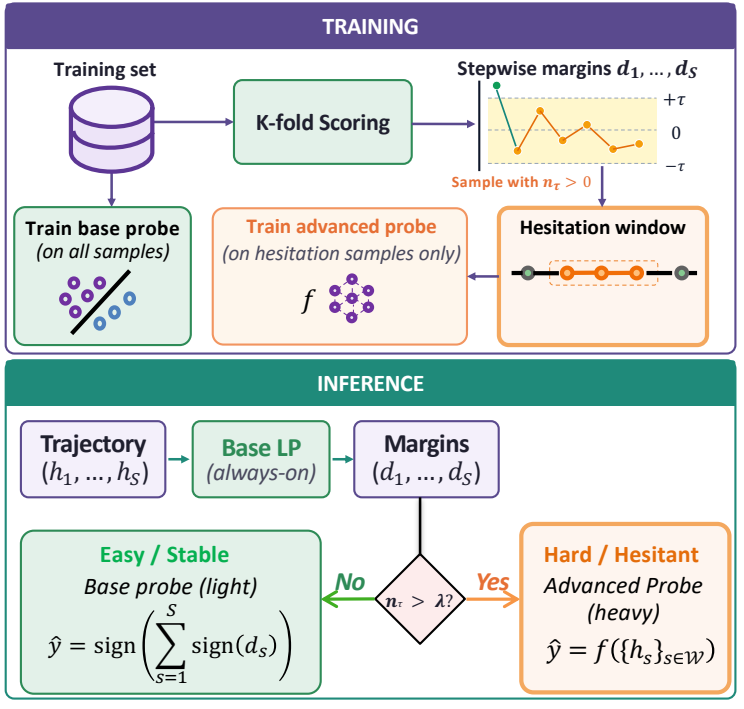
Hesitation step: $|d_s| < \tau$

Hesitation severity n_τ = Number of hesitation steps

Key finding: hesitation severity predicts when the lightweight probe is likely to fail.

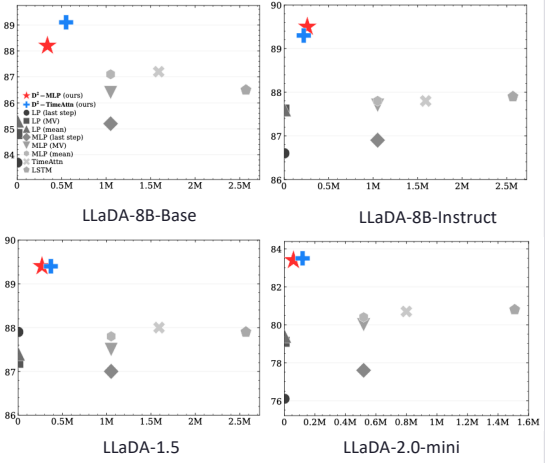
2 Solution: hesitation-aware training + routing

Use hesitation severity to identify hard trajectories, train heavy probe on them, and trigger it only when hesitation severity is sufficiently high at test time.



3 Effectiveness-efficiency tradeoffs

Expected test-time parameters vs F1. Red/blue points are D²-Monitor variants; gray points are 8 baselines.



D^2-Monitor sits top-left

Best trade-off between effectiveness (F1) and efficiency relative to 8 baselines.

4 D-LLMs | 3 datasets | ≤0.93M params